Rajinder Lal
Math 400

Math 400 Exam 2 Part 3 Due by Wednesday, October 21 11:59 pm

Instructions: Please show an appropriate amount of work for each problem. Write legibly and clearly. Please circle your answer. Make sure to simplify all answers.

You can print out this exam and write your solutions directly on the exam. If you are unable to print out the exam, then you can write your solutions on separate pieces of paper. If you write your solutions on separate paper, then **make sure to put the problems in numerical order**. **Please write your name on the exam.**

You'll have to submit Part 3 of the exam to me in **Canvas**. Please scan your exam and upload your **single PDF file** to Canvas. **Look under Assignments – Exams – Exam 2 Part 3.**

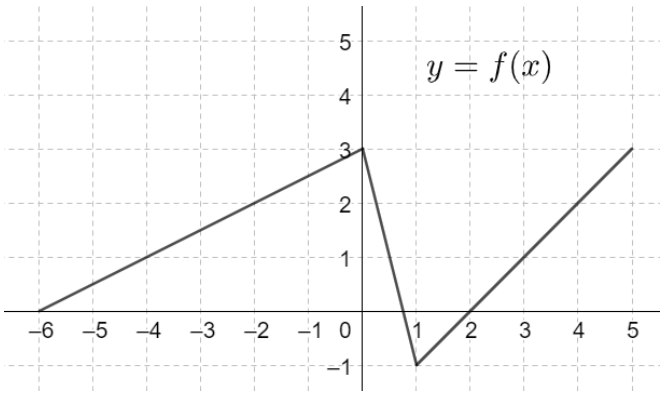
****This is an exam and so you shouldn't be working on the exam with other people and/or use apps/websites that can solve the problem for you.**

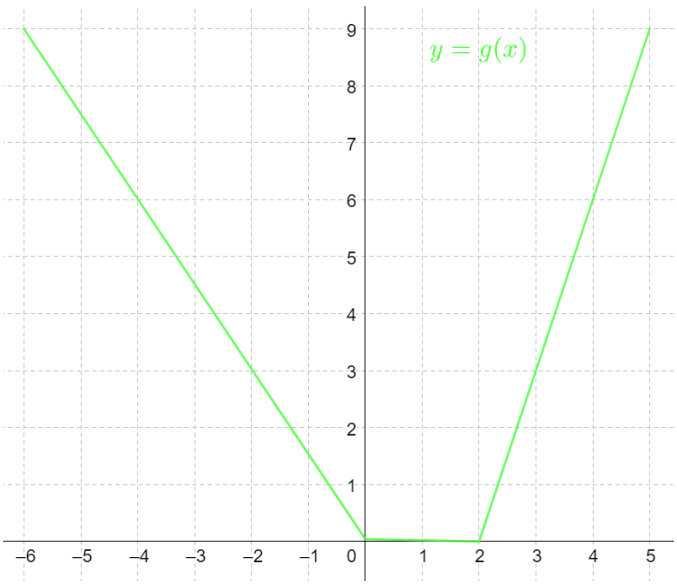
Name: _____

Math 400 Exam 2 Part 3 40 Total Points

1. The graphs of the function f and g are found below. Let $A(x) = f(x)g(x)$ and $B(x) = \frac{f(x)}{g(x)}$. Use the graphs of f and g to find the following values.

Use exact values. No decimal approximations.





(1.5 pts) $(a.)\ f'(2)$

(1.5 pts) $(b.)\ g'(1)$

(2.5 pts) $(c.)\ A'(-2)$

(2.5 pts) $(d.)\ B'(3)$

(8 pts) 2. Find the derivative of the function using the definition of a derivative,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

$f(x) = \sqrt{4x+1}$

(8 pts) 3. Find the values of x where the tangent line to the graph of f is horizontal. **DO NOT USE LOGARITHMIC DIFFERENTIATION.**

$$h(x) = (2x + 3)^7(8x - 9)^{12}$$

(8 pts) 4. Find the derivative of the following function. **DO NOT SIMPLIFY.**

$$y = \log_6 \sqrt[3]{\frac{3x^2+1}{x^4+2}}$$

(8 pts) 5. Find the derivative of the following function. **DO NOT SIMPLIFY.**

$$y = \csc^2[\tan(3^{4x} + \ln x)]$$